§2 偏 导 数

一、 偏导数的定义及其计算

二、高阶偏导数

三、小结 思考题

一、偏导数的定义及其计算法

定义　设函数$z=f(x,y)$在点(x_0,y_0)的某一邻域内有定义，当y固定在y_0而x在x_0处有增量Δx时，相应地函数有增量

$$f(x_0+\Delta x,y_0)-f(x_0,y_0),$$

如果$\lim\limits_{\Delta x\to 0}\dfrac{f(x_0+\Delta x,y_0)-f(x_0,y_0)}{\Delta x}$存在，则称此极限为函数$z=f(x,y)$在点$(x_0,y_0)$处对$x$的偏导数，记为

$$\left.\frac{\partial z}{\partial x}\right|_{\substack{x=x_0\\ y=y_0}},\quad \left.\frac{\partial f}{\partial x}\right|_{\substack{x=x_0\\ y=y_0}},\quad \left.z'_x\right|_{\substack{x=x_0\\ y=y_0}} \text{或 } f'_x(x_0,y_0).$$

同理可定义函数$z=f(x,y)$在点(x_0,y_0)处对y的偏导数，　为

$$\lim_{\Delta y\to 0}\frac{f(x_0,y_0+\Delta y)-f(x_0,y_0)}{\Delta y}$$

记为$\left.\dfrac{\partial z}{\partial y}\right|_{\substack{x=x_0\\ y=y_0}}$，$\left.\dfrac{\partial f}{\partial y}\right|_{\substack{x=x_0\\ y=y_0}}$，$\left.z'_y\right|_{\substack{x=x_0\\ y=y_0}}$或$f'_y(x_0,y_0)$.

如果函数$z=f(x,y)$在区域D内任一点(x,y)处对x的偏导数都存在，那么这个偏导数就是x、y的函数，它就称为函数$z=f(x,y)$对自变量x的偏导数，

记作$\dfrac{\partial z}{\partial x}$，$\dfrac{\partial f}{\partial x}$，$z'_x$ 或 $f'_x(x,y)$.

同理可以定义函数$z=f(x,y)$对自变量y的偏导数，

记作$\dfrac{\partial z}{\partial y}$，$\dfrac{\partial f}{\partial y}$，$z'_y$ 或 $f'_y(x,y)$.

偏导数的概念可以推广到二元以上函数

如 $u = f(x,y,z)$ 在 (x,y,z) 处

$$f'_x(x,y,z) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y, z) - f(x,y,z)}{\Delta x},$$

$$f'_y(x,y,z) = \lim_{\Delta y \to 0} \frac{f(x, y+\Delta y, z) - f(x,y,z)}{\Delta y},$$

$$f'_z(x,y,z) = \lim_{\Delta z \to 0} \frac{f(x, y, z+\Delta z) - f(x,y,z)}{\Delta z}.$$

例 1 求 $z=x^2+3xy+y^2$ 在点(1,2)处的偏导数.

解 $\dfrac{\partial z}{\partial x}=2x+3y$; $\dfrac{\partial z}{\partial y}=3x+2y$.

$$\therefore \left.\frac{\partial z}{\partial x}\right|_{\substack{x=1\\y=2}}=2\times1+3\times2=8,$$

$$\left.\frac{\partial z}{\partial y}\right|_{\substack{x=1\\y=2}}=3\times1+2\times2=7.$$

例 2 设 $z = x^y (x > 0, x \neq 1)$，

求证 $\dfrac{x}{y}\dfrac{\partial z}{\partial x} + \dfrac{1}{\ln x}\dfrac{\partial z}{\partial y} = 2z.$

证 $\dfrac{\partial z}{\partial x} = yx^{y-1}$, $\quad \dfrac{\partial z}{\partial y} = x^y \ln x$,

$$\frac{x}{y}\frac{\partial z}{\partial x} + \frac{1}{\ln x}\frac{\partial z}{\partial y} = \frac{x}{y} yx^{y-1} + \frac{1}{\ln x} x^y \ln x$$

$= x^y + x^y = 2z.$ 原结论成立.

例 3　设 $z=\arcsin\dfrac{x}{\sqrt{x^2+y^2}}$，求 $\dfrac{\partial z}{\partial x}$，$\dfrac{\partial z}{\partial y}$.

解

$$\frac{\partial z}{\partial x}=\frac{1}{\sqrt{1-\dfrac{x^2}{x^2+y^2}}}\cdot\left(\frac{x}{\sqrt{x^2+y^2}}\right)'_x$$

$$=\frac{\sqrt{x^2+y^2}}{|y|}\cdot\frac{y^2}{\sqrt{(x^2+y^2)^3}}\qquad(\sqrt{y^2}=|y|)$$

$$=\frac{|y|}{x^2+y^2}.$$

$$\frac{\partial z}{\partial y}=\frac{1}{\sqrt{1-\frac{x^2}{x^2+y^2}}}\cdot\left(\frac{x}{\sqrt{x^2+y^2}}\right)_y'$$

$$=\frac{\sqrt{x^2+y^2}}{|y|}\cdot\frac{(-xy)}{\sqrt{(x^2+y^2)^3}}$$

$$=-\frac{x}{x^2+y^2}\operatorname{sgn}\frac{1}{y}\qquad(y\neq 0)$$

$$\left.\frac{\partial z}{\partial y}\right|_{\substack{x\neq 0\\ y=0}}\text{不存在.}$$

有关偏导数的几点说明：

1、 偏导数$\dfrac{\partial u}{\partial x}$是一个整体记号，不能拆分；

2、 求分界点、不连续点处的偏导数要用定义求；

例如，设$z = f(x, y) = \sqrt{|xy|}$，求$f'_x(0, 0)$，$f'_y(0, 0)$.

解　$$f'_x(0,0) = \lim_{x\to 0}\frac{\sqrt{|x\cdot 0|}-0}{x} = 0 = f'_y(0,0).$$

例 5 设 $f(x,y)=\begin{cases}\dfrac{xy}{x^2+y^2} & (x,y)\neq(0,0)\\ 0 & (x,y)=(0,0)\end{cases}$

求 $f(x,y)$的偏导数.

解 当$(x,y)\neq(0,0)$时,

$$f'_x(x,y)=\frac{y(x^2+y^2)-2x\cdot xy}{(x^2+y^2)^2}=\frac{y(y^2-x^2)}{(x^2+y^2)^2},$$

$$f'_y(x,y)=\frac{x(x^2+y^2)-2y\cdot xy}{(x^2+y^2)^2}=\frac{x(x^2-y^2)}{(x^2+y^2)^2},$$

当$(x,y)=(0,0)$时，按定义可知：

$$f_x'(0,0)=\lim_{\Delta x\to 0}\frac{f(\Delta x,0)-f(0,0)}{\Delta x}=\lim_{\Delta x\to 0}\frac{0}{\Delta x}=0,$$

$$f_y'(0,0)=\lim_{\Delta y\to 0}\frac{f(0,\Delta y)-f(0,0)}{\Delta y}=\lim_{\Delta y\to 0}\frac{0}{\Delta y}=0,$$

$$f_x'(x,y)=\begin{cases}\dfrac{y(y^2-x^2)}{(x^2+y^2)^2} & (x,y)\neq(0,0)\\ 0 & (x,y)=(0,0)\end{cases},$$

$$f_y'(x,y)=\begin{cases}\dfrac{x(x^2-y^2)}{(x^2+y^2)^2} & (x,y)\neq(0,0)\\ 0 & (x,y)=(0,0)\end{cases}.$$

3、偏导数存在与连续的关系

一元函数中在某点可导 $\longrightarrow$ 连续,

多元函数中在某点偏导数存在 $\overset{?}{\longrightarrow}$ 连续,

例如,函数 $f(x,y)=\begin{cases}\dfrac{xy}{x^2+y^2}, & x^2+y^2\neq 0\\ 0, & x^2+y^2=0\end{cases}$,

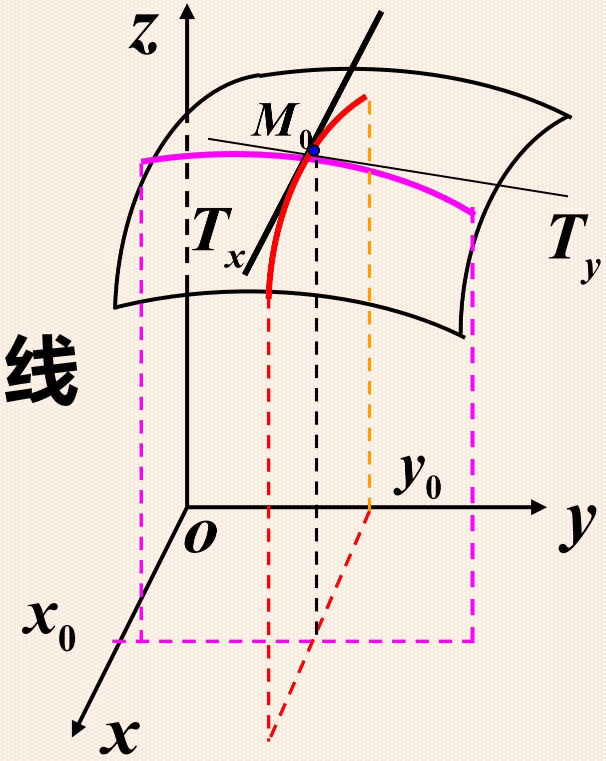
依定义知在 $(0,0)$ 处, $f_x'(0,0)=f_y'(0,0)=0$.

但函数在该点处并不连续. 偏导数存在 $\not\longrightarrow$ 连续.

4. 二元函数偏导数的几何意义:

$$\left.\frac{\partial f}{\partial x}\right|_{\substack{x=x_0\\y=y_0}} = \frac{\mathrm{d}}{\mathrm{d}x} f(x, y_0)\Big|_{x=x_0}$$

是曲线 $\begin{cases} z = f(x, y) \\ y = y_0 \end{cases}$ 在点 M_0 处的切线 $\mathrm{M_0T_x}$ 对 x 轴的斜率.



$$\left.\frac{\partial f}{\partial y}\right|_{\substack{x=x_0\\y=y_0}} = \frac{\mathrm{d}}{\mathrm{d}y} f(x_0, y)\Big|_{y=y_0}$$

是曲线 $\begin{cases} z = f(x, y) \\ x = x_0 \end{cases}$ 在点M_0 处的切线 $\mathrm{M_0T_y}$ 对 y 轴的斜率.

二、高阶偏导数

函数$z = f(x, y)$的二阶偏导数为

$$\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial x^2} = f''_{xx}(x, y), \quad \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial y^2} = f''_{yy}(x, y)$$

纯偏导

$$\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial x \partial y} = f''_{xy}(x, y), \quad \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial y \partial x} = f''_{yx}(x, y)$$

混合偏导

定义：二阶及二阶以上的偏导数统称为高阶偏导数.

例 6 设$z = x^3y^2 - 3xy^3 - xy + 1$，

求$\frac{\partial^2 z}{\partial x^2}$、$\frac{\partial^2 z}{\partial y \partial x}$、$\frac{\partial^2 z}{\partial x \partial y}$、$\frac{\partial^2 z}{\partial y^2}$及$\frac{\partial^3 z}{\partial x^3}$.

解 $\frac{\partial z}{\partial x} = 3x^2y^2 - 3y^3 - y$, $\frac{\partial z}{\partial y} = 2x^3y - 9xy^2 - x$;

$\frac{\partial^2 z}{\partial x^2} = 6xy^2$, $\frac{\partial^3 z}{\partial x^3} = 6y^2$, $\frac{\partial^2 z}{\partial y^2} = 2x^3 - 18xy$;

$\frac{\partial^2 z}{\partial x \partial y} = 6x^2y - 9y^2 - 1$, $\frac{\partial^2 z}{\partial y \partial x} = 6x^2y - 9y^2 - 1$.

例 7 设$u=e^{ax}\cos by$，求二阶偏导数.

解 $\dfrac{\partial u}{\partial x}=ae^{ax}\cos by$, $\dfrac{\partial u}{\partial y}=-be^{ax}\sin by$;

$$\frac{\partial^2 u}{\partial x^2}=a^2e^{ax}\cos by,\qquad \frac{\partial^2 u}{\partial y^2}=-b^2e^{ax}\cos by,$$

$$\frac{\partial^2 u}{\partial x\partial y}=-abe^{ax}\sin by,\qquad \frac{\partial^2 u}{\partial y\partial x}=-abe^{ax}\sin by.$$

问题： 混合偏导数都相等吗？

例 8 设 $f(x,y)=\begin{cases}\dfrac{x^3y}{x^2+y^2} & (x,y)\neq(0,0)\\ 0 & (x,y)=(0,0)\end{cases}$

求 $f(x,y)$在$(0,0)$的二阶混合偏导数.

解 当$(x,y)\neq(0,0)$时,

$$f'_x(x,y)=\frac{3x^2y(x^2+y^2)-2x\cdot x^3y}{(x^2+y^2)^2}=\frac{3x^2y}{x^2+y^2}-\frac{2x^4y}{(x^2+y^2)^2},$$

$$f'_y(x,y)=\frac{x^3}{x^2+y^2}-\frac{2x^3y^2}{(x^2+y^2)^2},$$

$(x,y)\neq(0,0)$时，

$$f'_x(x,y)=\frac{3x^2y}{x^2+y^2}-\frac{2x^4y}{(x^2+y^2)^2}$$

$$f'_y(x,y)=\frac{x^3}{x^2+y^2}-\frac{2x^3y^2}{(x^2+y^2)^2}$$

当$(x,y)=(0,0)$时，按定义可知：

$$f'_x(0,0)=\lim_{\Delta x\to 0}\frac{f(\Delta x,0)-f(0,0)}{\Delta x}=\lim_{\Delta x\to 0}\frac{0}{\Delta x}=0,$$

$$f'_y(0,0)=\lim_{\Delta y\to 0}\frac{f(0,\Delta y)-f(0,0)}{\Delta y}=\lim_{\Delta y\to 0}\frac{0}{\Delta y}=0,$$

$$f''_{xy}(0,0)=\lim_{\Delta y\to 0}\frac{f'_x(0,\Delta y)-f'_x(0,0)}{\Delta y}=0,$$

$$f''_{yx}(0,0)=\lim_{\Delta x\to 0}\frac{f'_y(\Delta x,0)-f'_y(0,0)}{\Delta x}=1.$$

显然 $f''_{xy}(0,0)\neq f''_{yx}(0,0).$

问题：具备怎样的条件才能使混合偏导数相等？

定理 如果函数$z=f(x,y)$的两个二阶混合偏导数$\dfrac{\partial^2 z}{\partial y\partial x}$及$\dfrac{\partial^2 z}{\partial x\partial y}$在区域 D 内连续，那么在该区域内这两个二阶混合偏导数必相等.

例 9 验证函数$u(x,y)=\ln\sqrt{x^2+y^2}$满足拉普拉斯方程 $\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}=0.$

解 $\because \ln\sqrt{x^2+y^2}=\dfrac{1}{2}\ln(x^2+y^2),$

$$\therefore \frac{\partial u}{\partial x} = \frac{x}{x^2 + y^2}, \quad \frac{\partial u}{\partial y} = \frac{y}{x^2 + y^2},$$

$$\therefore \frac{\partial^2 u}{\partial x^2} = \frac{(x^2 + y^2) - x \cdot 2x}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2},$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{(x^2 + y^2) - y \cdot 2y}{(x^2 + y^2)^2} = \frac{x^2 - y^2}{(x^2 + y^2)^2}.$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2} + \frac{x^2 - y^2}{(x^2 + y^2)^2} = 0.$$ 证毕.

三、小结

偏导数的定义 （偏增量比的极限）

偏导数的计算、偏导数的几何意义

$$\text{高阶偏导数}\begin{cases}\text{纯偏导}\\ \text{混合偏导（相等的条件）}\end{cases}$$

1. 偏导数的概念及有关结论

- 定义; 记号; 几何意义
- 函数在一点偏导数存在 $\not\Longrightarrow$ 函数在此点连续
- 混合偏导数连续 $\Longrightarrow$ 与求导顺序无关

2. 偏导数的计算方法

- 求一点处偏导数的方法
  - 先代后求
  - 先求后代
  - 利用定义
- 求高阶偏导数的方法 —— 逐次求导法

(与求导顺序无关时, 应选择方便的求导顺序)